Computer Communication & Networks

Lecture 22

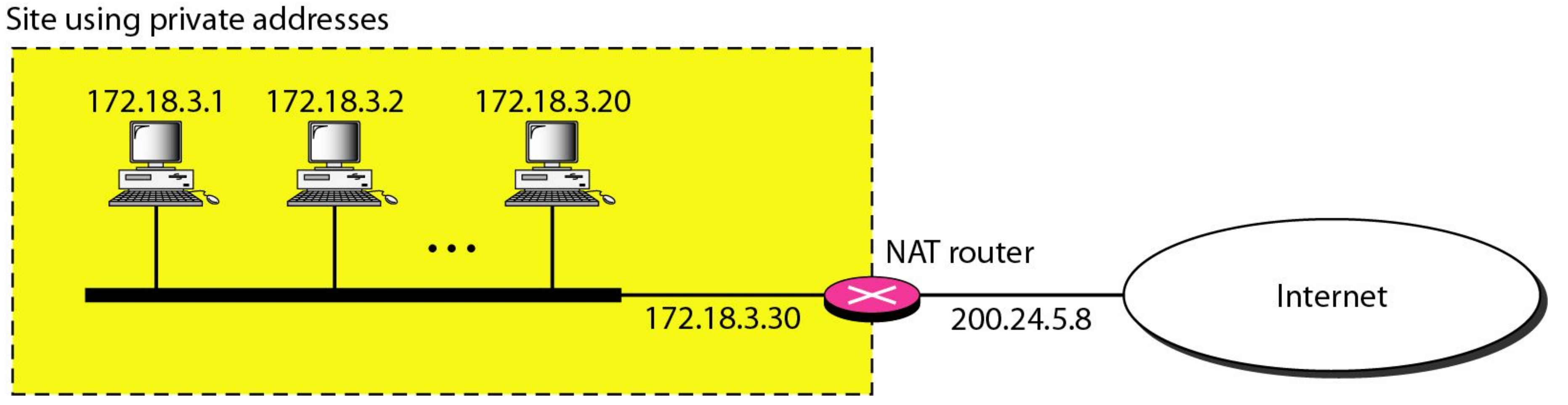
Network Layer: IP and Address Mapping

Network Address Translation (NAT)

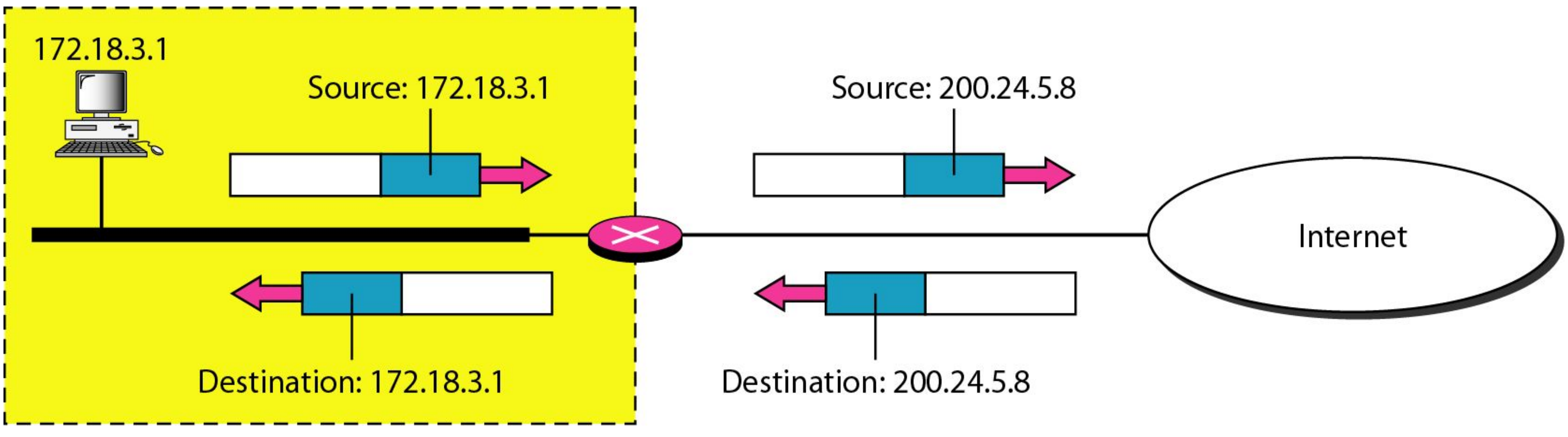
Addresses for private networks

Range			*Total*
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

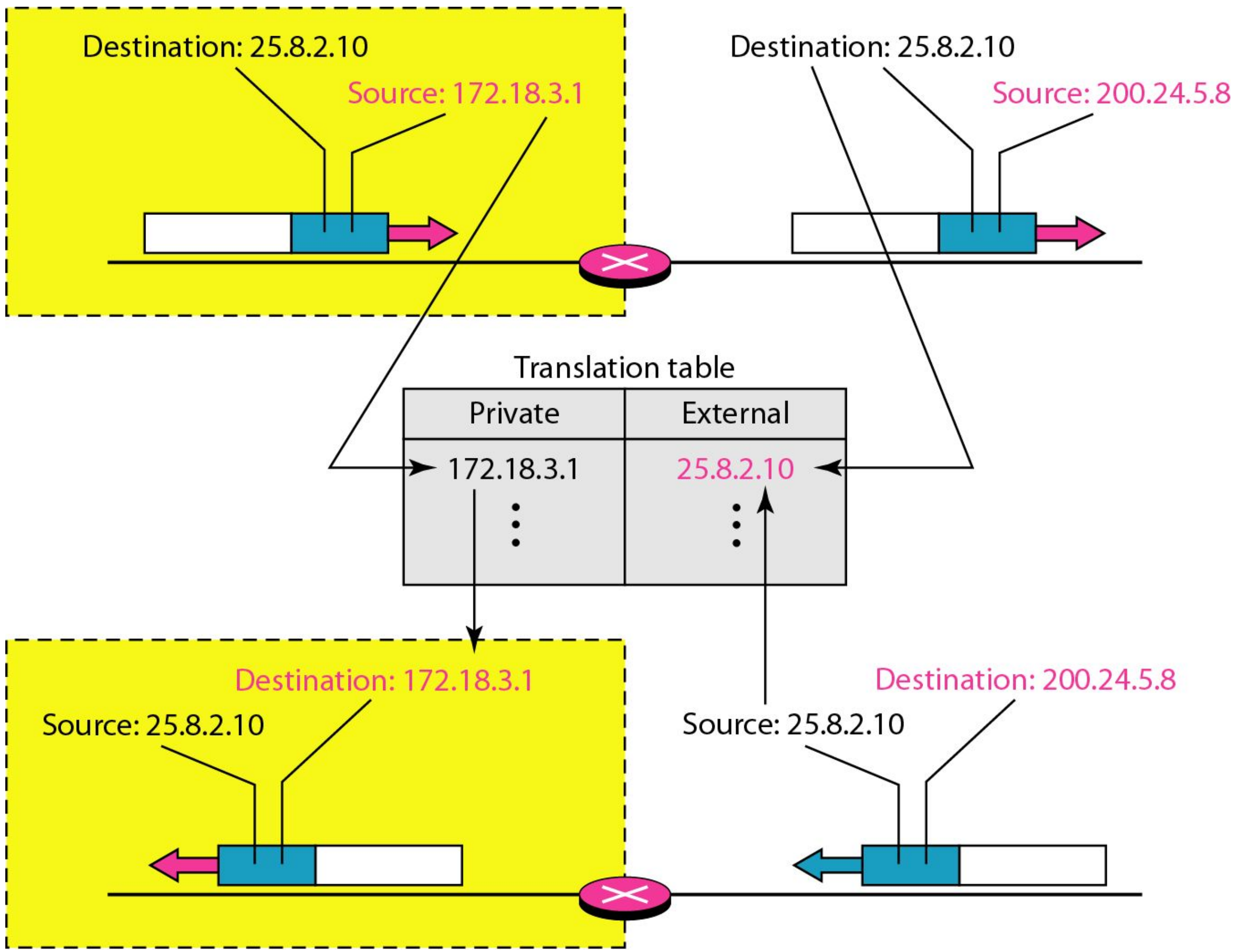
A NAT Implementation



Addresses in a NAT



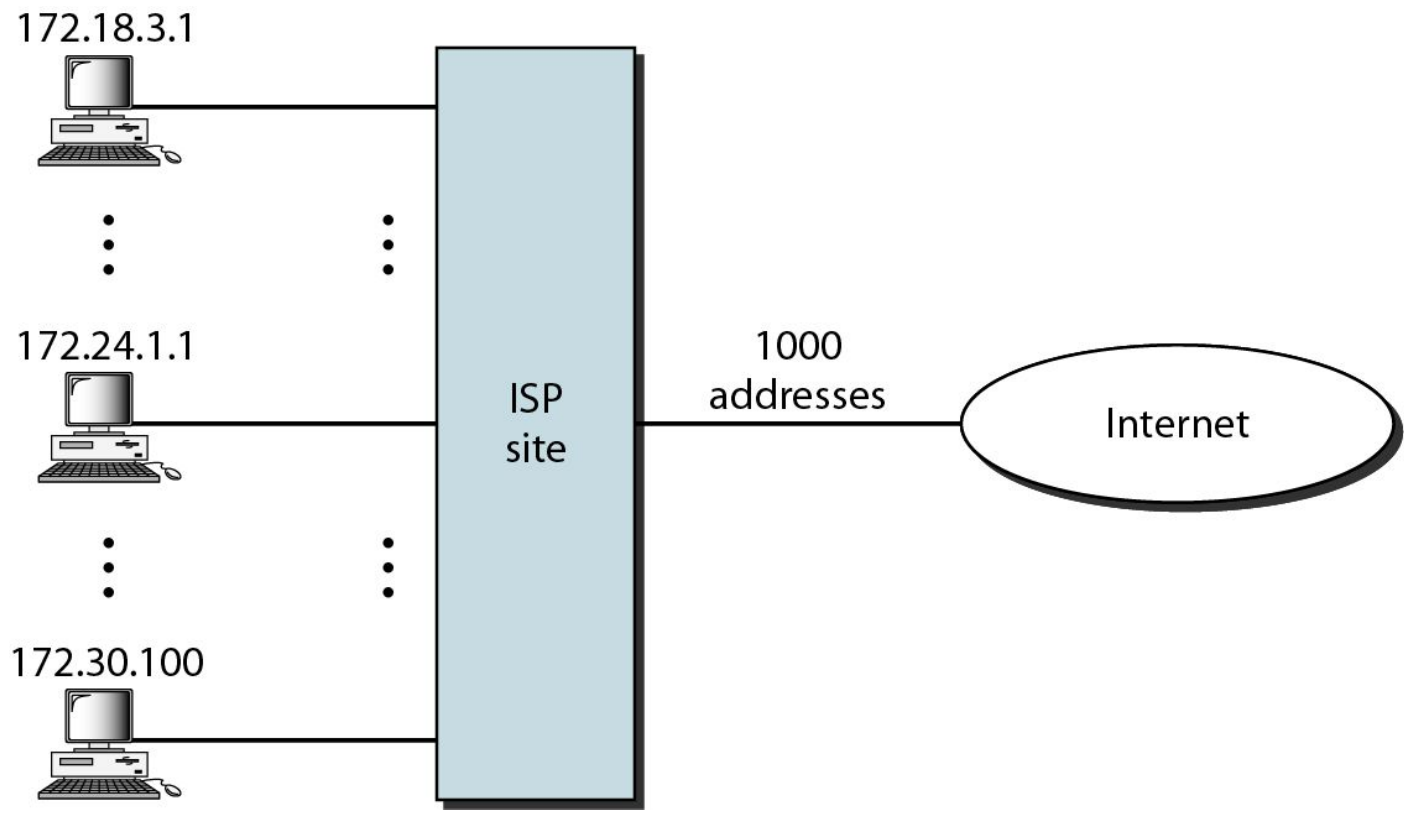
NAT Address Translation



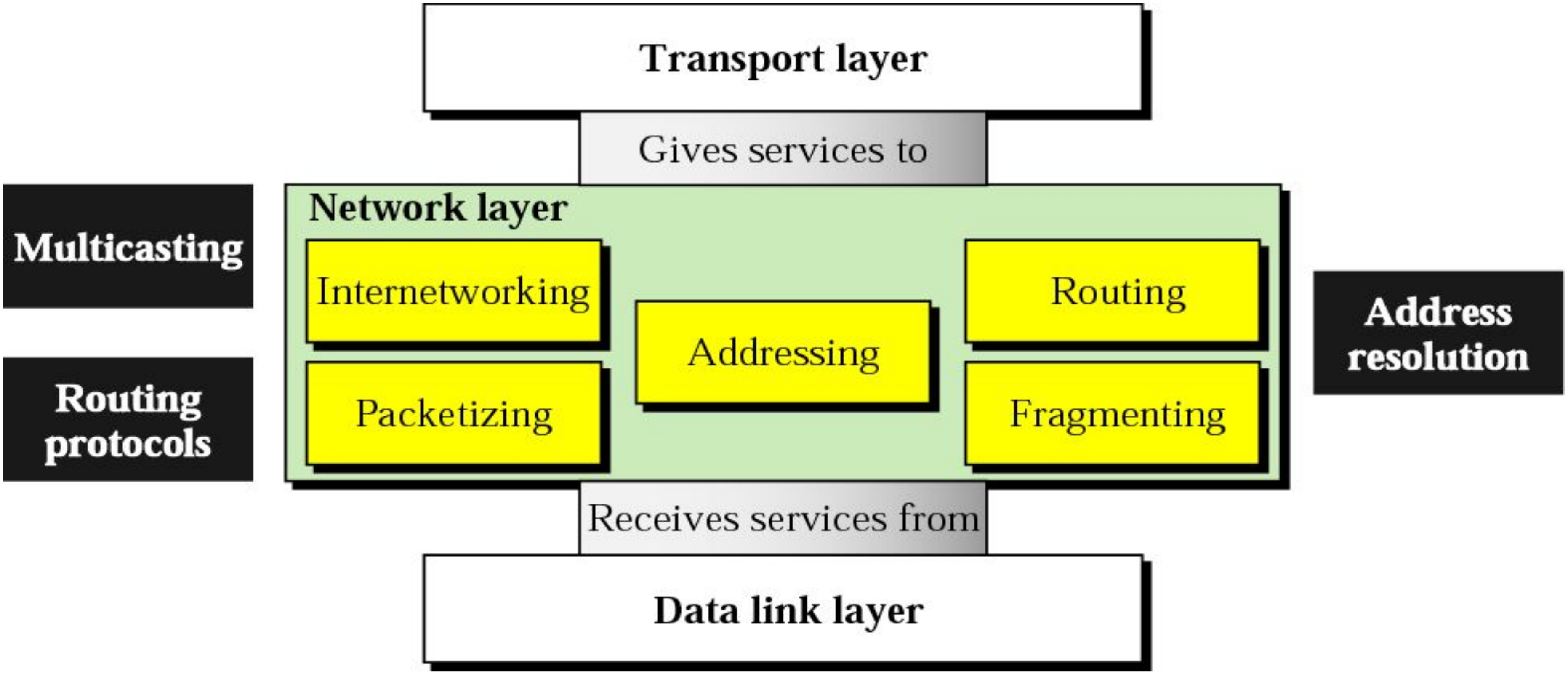
Five-column translation table

Private Address	*Private Port*	*External Address*	*External Port*	*Transport Protocol*
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
. . .	. . .	. . .	. . .	. . .

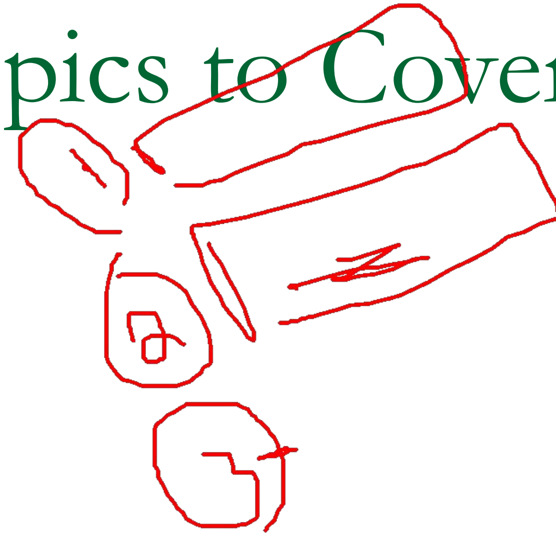
An ISP and NAT



Network Layer



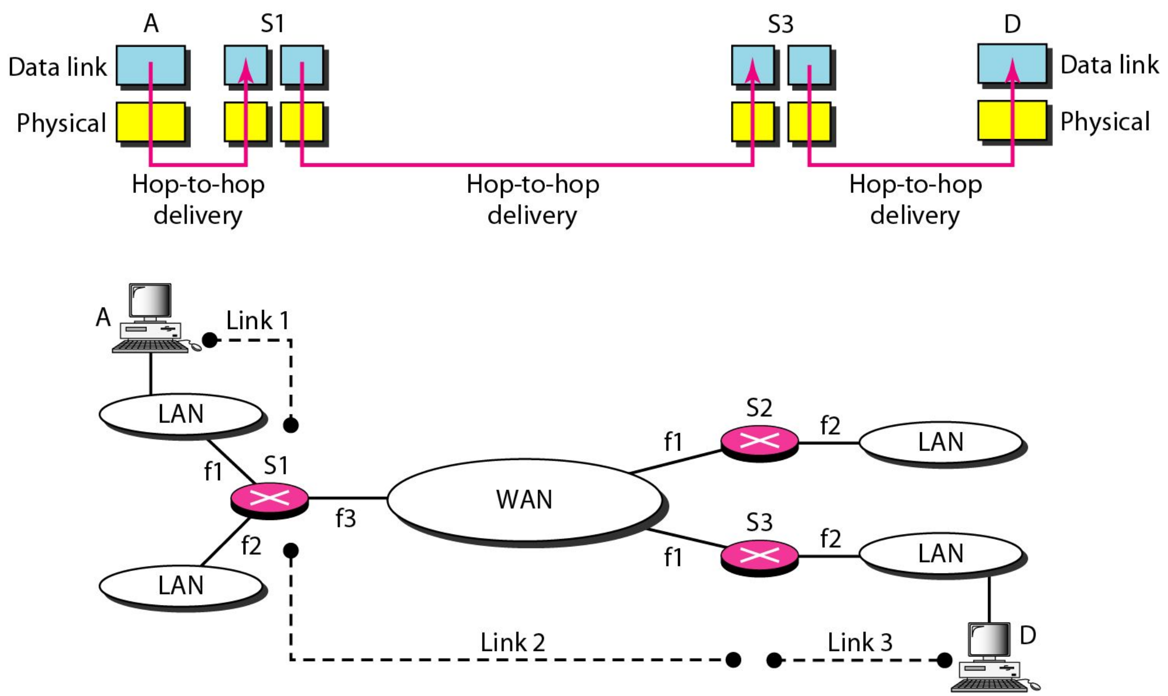
Network Layer Topics to Cover



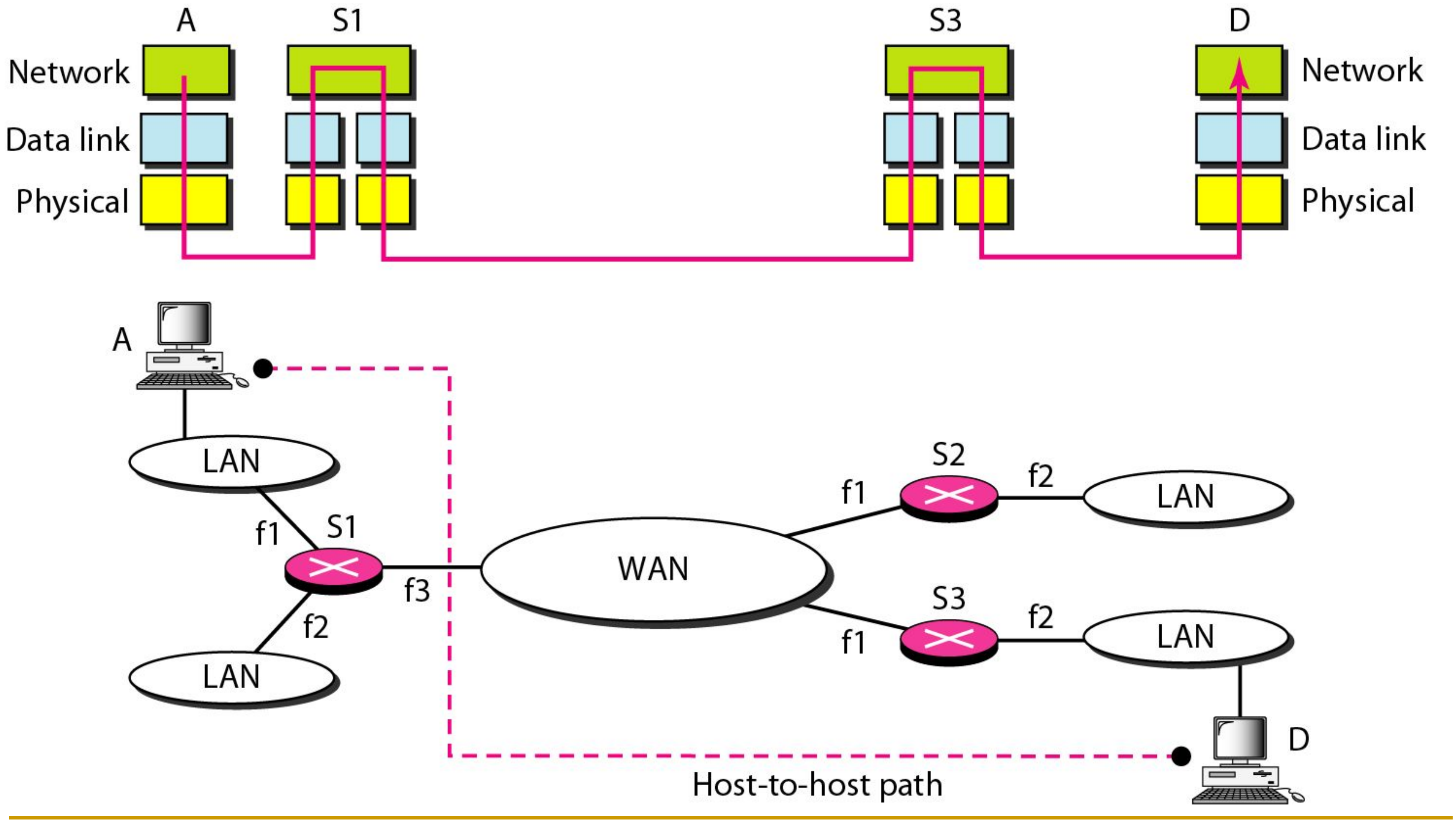
- *Logical Addressing*
- ***Internet Protocol***
- *Address Mapping*
- *Delivery, Forwarding, Routing*

Internetworking

- In this section, we discuss internetworking, connecting networks together to make an internetwork or an internet.

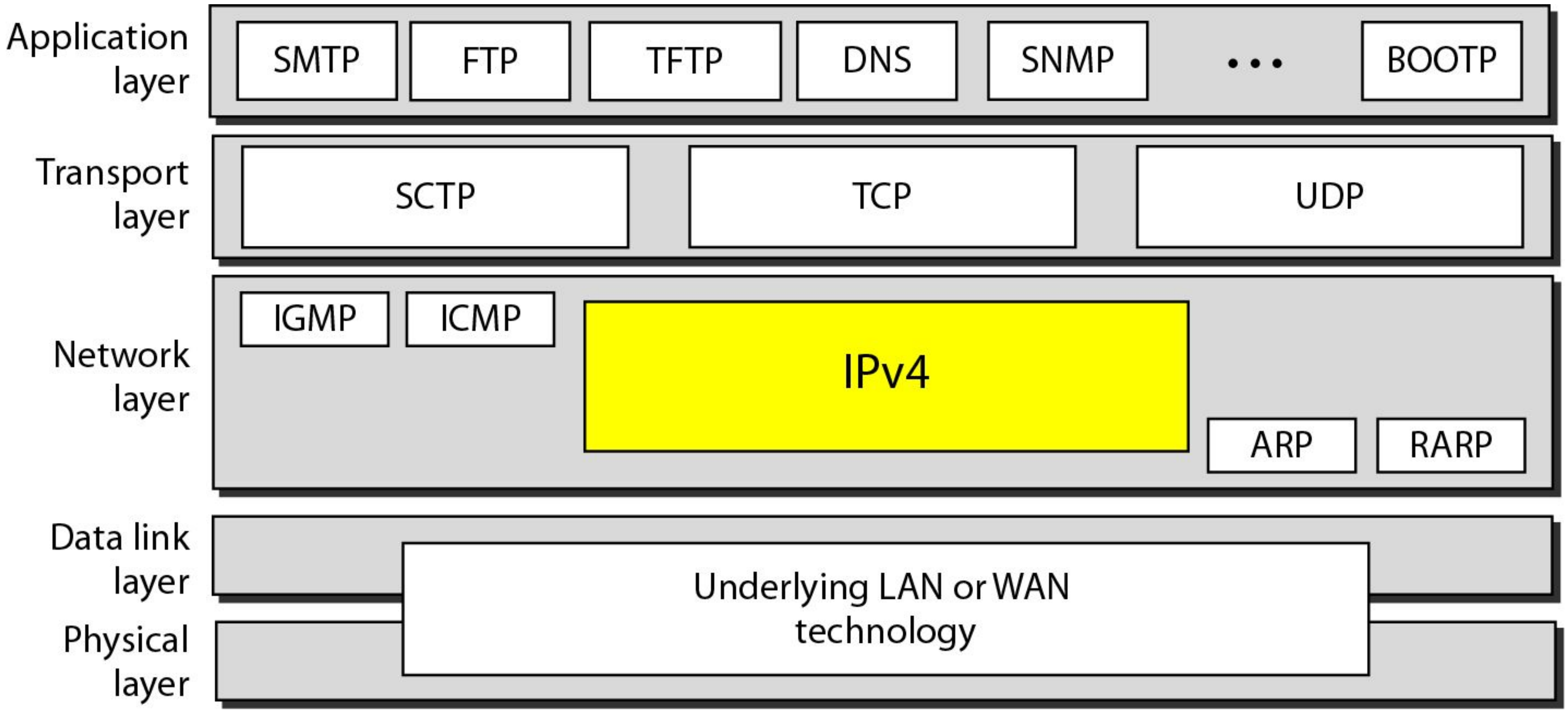


Network layer in an Internetwork

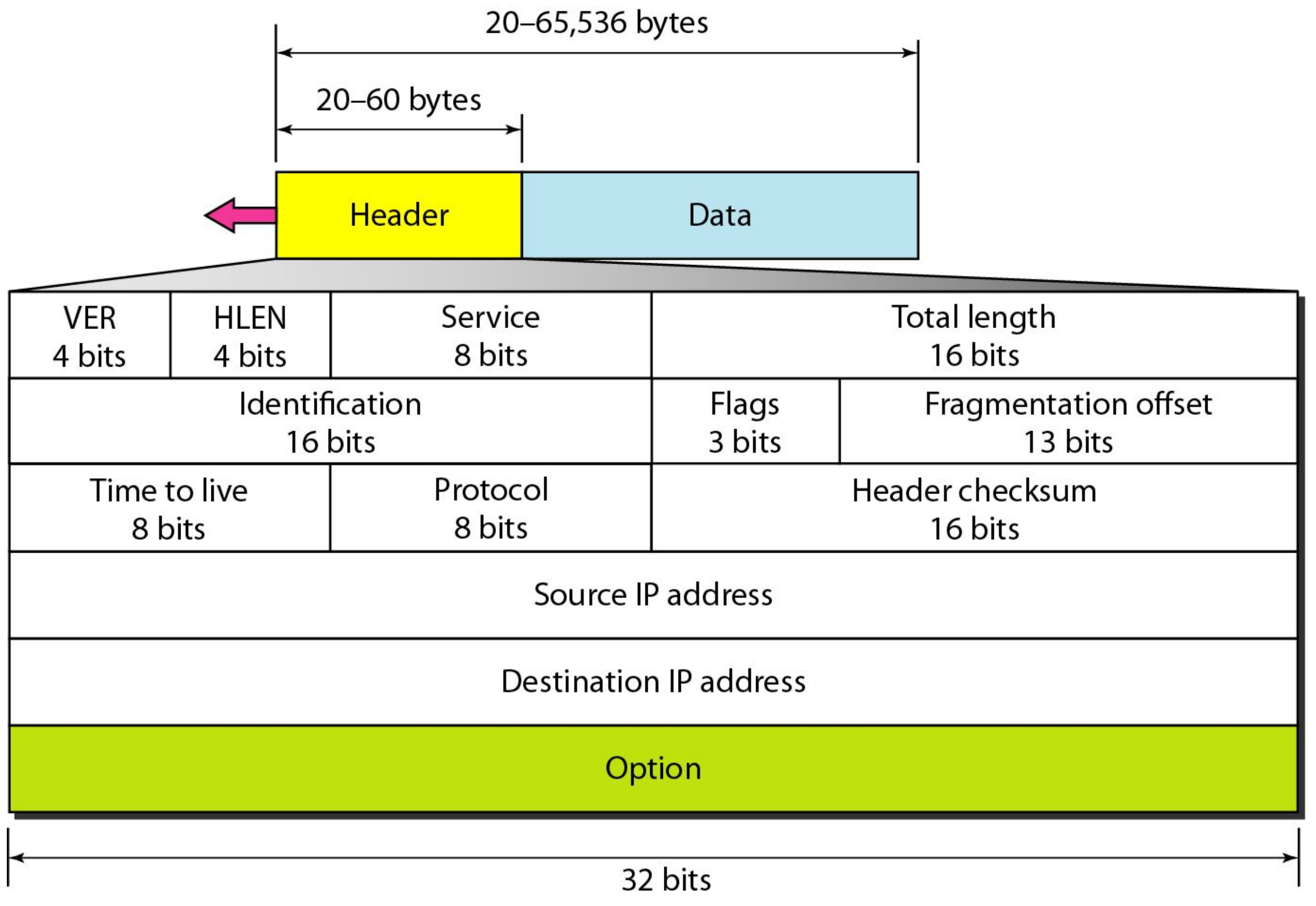


IPv4

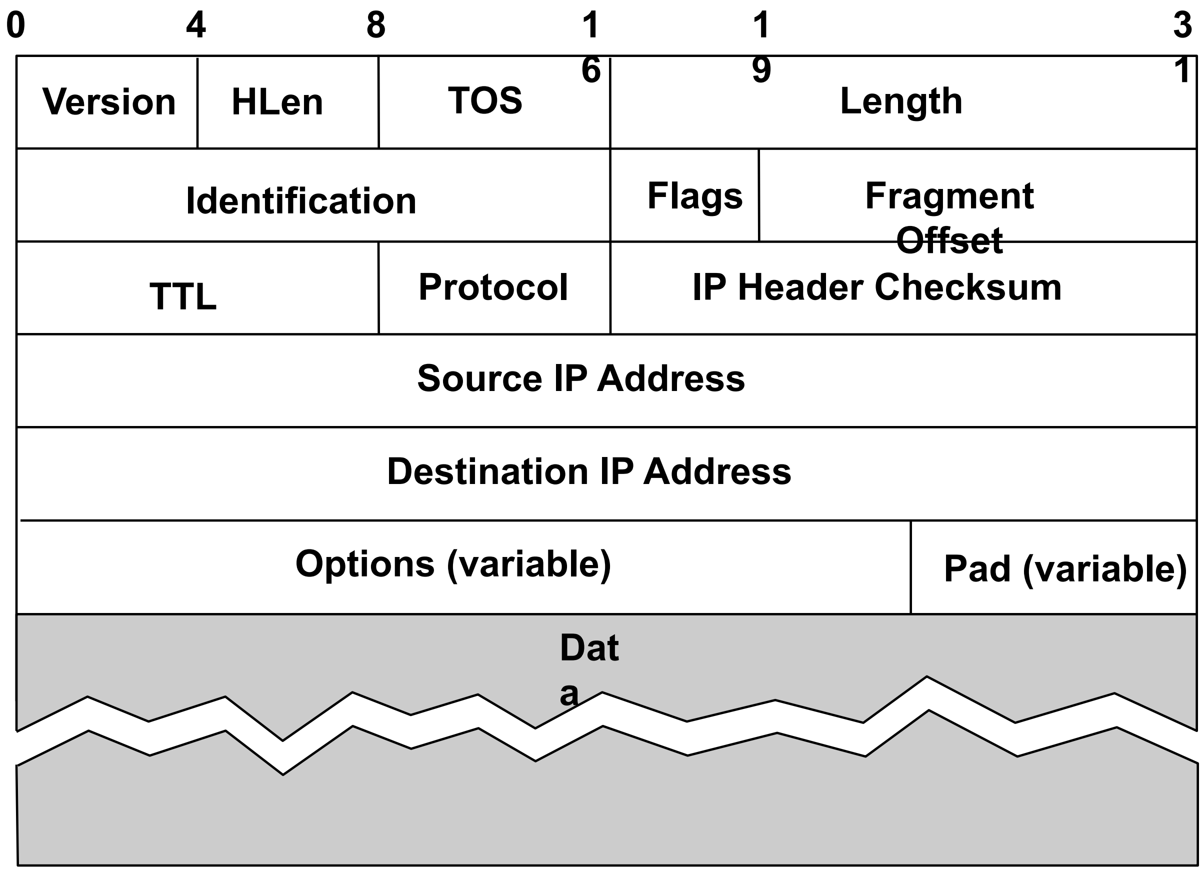
- The Internet Protocol version 4 (IPv4) is the delivery mechanism used by the TCP/IP protocols.



IPv4 datagram format



IP Packet Format



Current IP Protocol Version is 4, called IPv4

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
TTL		Protocol	IP Header Checksum		
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

Length of IP Header in number of 32 bit words including options. Maximum header size is 60 bytes.

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
TTL		Protocol	IP Header Checksum		
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

The type-of-service field is composed of a 3-bit precedence field.
(Which are largely ignored in current routers).
4 TOS bits and an unused bit that must be zero.

<table>
<tr><td>0</td><td>4</td><td>8</td><td>16</td><td>19</td><td>31</td></tr>
<tr><td>Version</td><td>HLen</td><td colspan="2">TOS</td><td colspan="2">Length</td></tr>
<tr><td colspan="4">Identification</td><td>Flags</td><td>Fragment Offset</td></tr>
<tr><td colspan="3">TTL</td><td>Protocol</td><td colspan="2">IP Header Checksum</td></tr>
<tr><td colspan="6">Source IP Address</td></tr>
<tr><td colspan="6">Destination IP Address</td></tr>
<tr><td colspan="5">Options (variable)</td><td>Pad (variable)</td></tr>
</table>

4 TOS bits are: minimize delay, maximize throughput, maximize reliability, and minimize monetary cost. Only one of these bits can be turned on. All 4 bits set to 0 means normal service.

0 4 8 16 19 31

Version	HLen	TOS	Length	
Identification			Flags	Fragment Offset
TTL		Protocol	IP Header Checksum	
Source IP Address				
Destination IP Address				
Options (variable)				Pad (variable)

Total length of IP datagram in bytes. It is a 16 bit field. Largest size of an IP datagram is 65635 bytes. Maximum header size is 60 bytes. Link layer MTU may restrict this size further.

0 4 8 16 19 31

Version	HLen	TOS	Length	
Identification			Flags	Fragment Offset
TTL		Protocol	IP Header Checksum	
Source IP Address				
Destination IP Address				
Options (variable)				Pad (variable)

Identification field uniquely identifies each datagram sent by a host. It is normally incremented by one each time a host sends a datagram. Very useful for fragmentation and reassembly.

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
TTL		Protocol	IP Header Checksum		
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

flags field also used for fragmentation and reassembly.

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
TTL		Protocol	IP Header Checksum		
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

Fragmentation offset used for fragmentation and reassembly.

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<tr><td>0</td><td>4</td><td>8</td><td>16</td><td>19</td><td></td><td>31</td></tr>
<tr><td>Version</td><td>HLen</td><td>TOS</td><td colspan="4">Length</td></tr>
<tr><td colspan="3">Identification</td><td>Flags</td><td colspan="3">Fragment Offset</td></tr>
<tr><td colspan="2">TTL</td><td>Protocol</td><td colspan="4">IP Header Checksum</td></tr>
<tr><td colspan="7">Source IP Address</td></tr>
<tr><td colspan="7">Destination IP Address</td></tr>
<tr><td colspan="5">Options (variable)</td><td colspan="2">Pad (variable)</td></tr>
</table>

Time-to-live or TTL field sets an upper limit on how man routers a datagram can go through. Every router decrements TTL by 1 before sending it forward. If TTL reaches 0 the datagram is dropped and an ICMP message is sent to the host application.

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
TTL		Protocol	IP Header Checksum		
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

Identifies the protocol that sent the datagram. The protocol (today) can be ICMP, IGMP, TCP, UDP

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
TTL		Protocol	IP Header Checksum		
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

The header checksum is calculated over the IP header only. TCP, UDP etc protect their own data and header by a checksum.

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<tr><th>0</th><th>4</th><th>8</th><th>16</th><th>19</th><th>31</th></tr>
<tr><td>Version</td><td>HLen</td><td>TOS</td><td colspan="3">Length</td></tr>
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0	4	8	16	19 31
Version	HLen	TOS	Length	
Identification			Flags	Fragment Offset
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Source IP Address				
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Options (variable)				Pad (variable)

- **Security handling used for military purposes (remember ARPANET was funded by US Defense),**

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
TTL		Protocol	IP Header Checksum		
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

- **Security handling used for military purposes (remember ARPANET was funded by US Defense),**
- **record route (each router on the way adds its address),**

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
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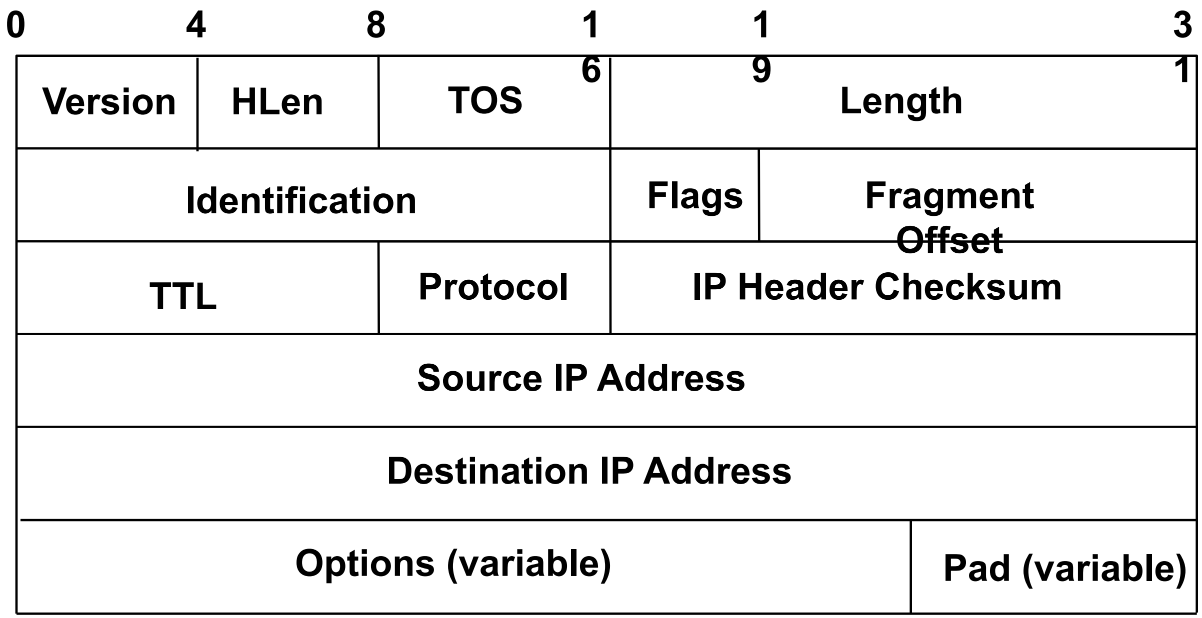
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

- Security handling used for military purposes (remember ARPANET was funded by US Defense),
- record route (each router on the way adds its address),
- time stamp (each router on the way adds its address and time stamp),

0	4	8	16	19	31
Version	HLen	TOS	Length		
Identification			Flags	Fragment Offset	
TTL		Protocol	IP Header Checksum		
Source IP Address					
Destination IP Address					
Options (variable)				Pad (variable)	

- Security handling used for military purposes (remember ARPANET was funded by US Defense),
- record route (each router on the way adds its address),
- time stamp (each router on the way adds its address and time stamp),
- loose source routing, strict source routing.



Options field always ends at a 32 bit boundary. Padding added as needed.

Example

An IPv4 packet has arrived with the first 8 bits as shown:

01000010

The receiver discards the packet. Why?

Solution

There is an error in this packet. The 4 leftmost bits (0100) show the version, which is correct. The next 4 bits (0010) show an invalid header length ($2 \times 4 = 8$). The minimum number of bytes in the header must be 20. The packet has been corrupted in transmission.

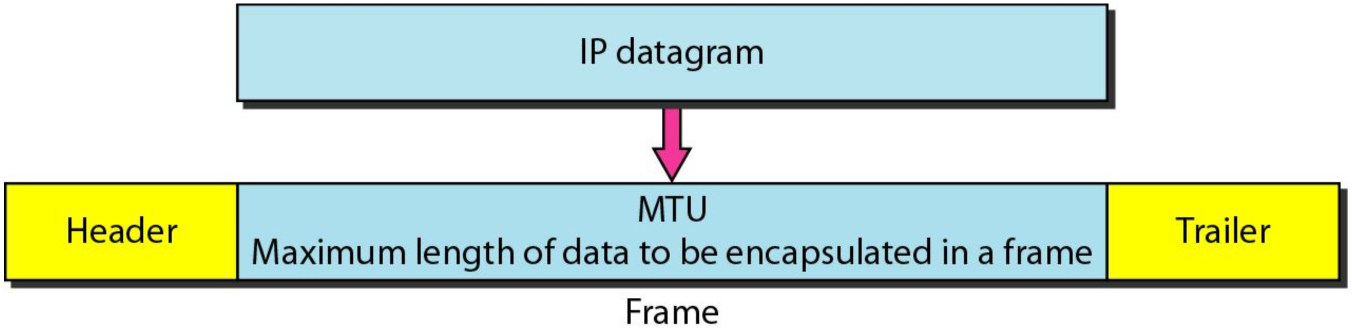
Example

In an IPv4 packet, the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?

Solution

The HLEN value is 8, which means the total number of bytes in the header is 8 × 4, or 32 bytes. The first 20 bytes are the base header, the next 12 bytes are the options.

Maximum transfer unit (MTU)



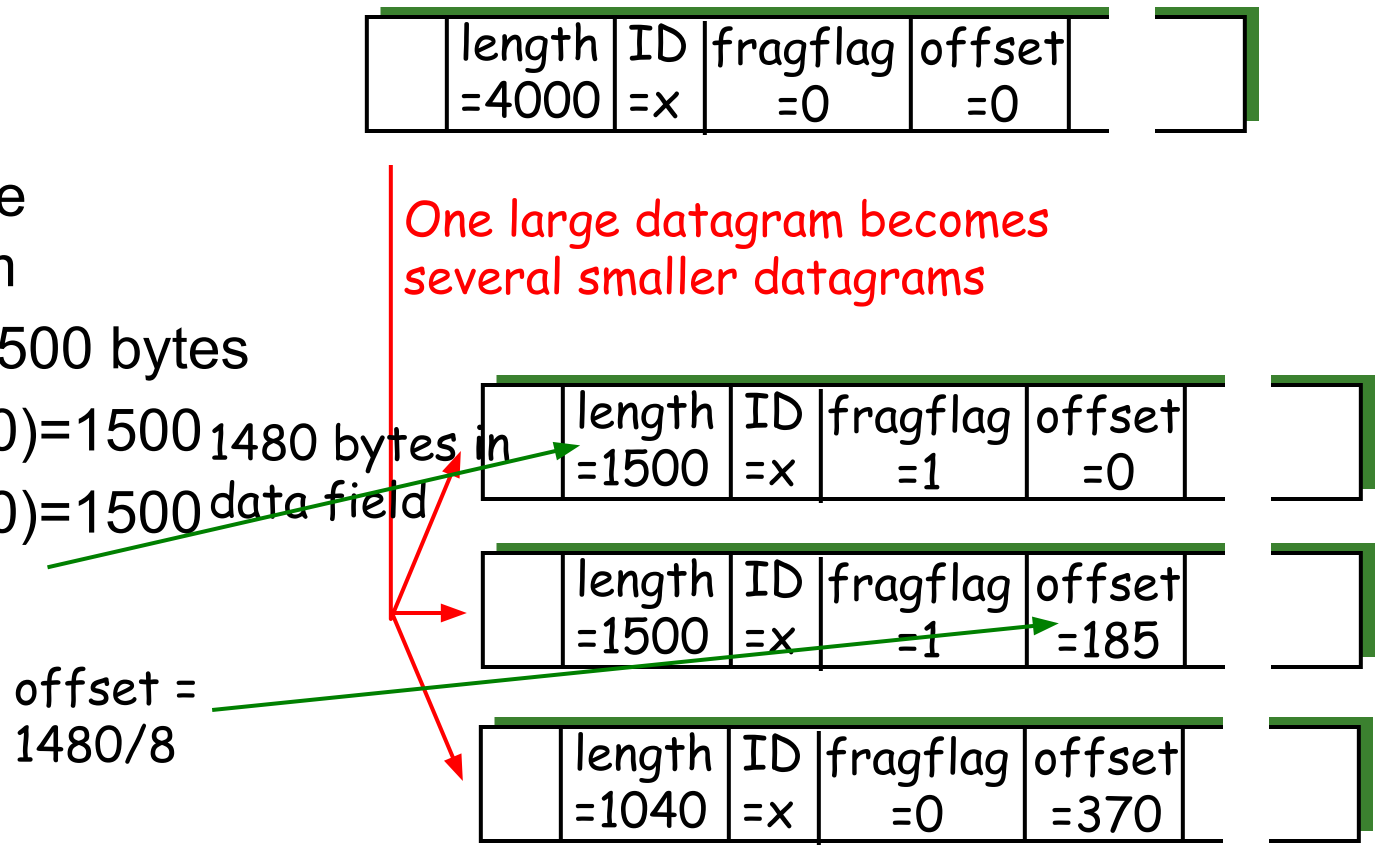
Protocol	*MTU*
Hyperchannel	65,535
Token Ring (16 Mbps)	17,914
Token Ring (4 Mbps)	4,464
FDDI	4,352
Ethernet	1,500
X.25	576
PPP	296

MTUs for some networks

IP Fragmentation and Reassembly

Example

- 4000 byte datagram
- MTU = 1500 bytes
- (1480+20)=1500
- (1480+20)=1500
- +1480



Example: Fragmenting a Packet

- A packet is to be forwarded to a network with MTU of 576 bytes. The packet has an IP header of 20 bytes and a data part of 1484 bytes. and of each fragment.
- Maximum data length per fragment = 576 - 20 = 556 bytes.
- We set maximum data length to 552 bytes to get multiple of 8.

	Total Length	Id	MF	Fragment Offset
Original packet	1504	x	0	0
Fragment 1	572	x	1	0
Fragment 2	572	x	1	69
Fragment 3	400	x	0	138